AI/NLP Risk Engine

From unstructured news to tactical index rebalancing and portfolio stress testing

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Problem & Approach

THE PROBLEM

Market-moving news and social posts arrive as unstructured text, faster than analysts can read.

Index desks need to react in minutes; risk teams need to know when a bank's book is exposed.

Both need the same machine-readable signal.

OUR APPROACH

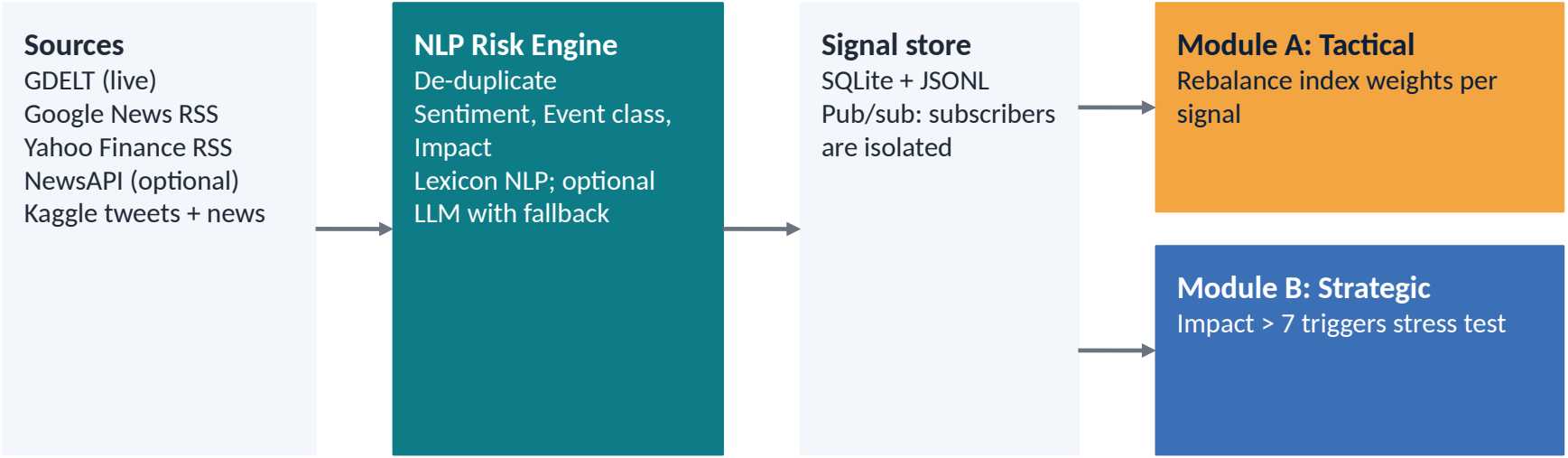
1. **Ingest**
   GDELT, Google News RSS, Yahoo RSS, Kaggle tweets and news

2. **Analyse**
   NLP turns each item into Sentiment (-1..+1), Event Class and Impact (1-10)

3. **Publish**
   One signal stream feeds both downstream modules

4. **Act**
   A: rebalance a mock S&P 100 index. B: stress-test a bank's loan book

System Design



The engine never imports its consumers: modules subscribe, and a failing subscriber cannot stop the feed.

Implementation Highlights

Signal engine

Rule/lexicon NLP: deterministic, offline, no API key. Optional OpenAI/Anthropic backend falls back to local on any failure.

Why: reproducible and free by default

Module A: Index

Impact-scaled weight moves under bounds and a turnover cap. Back-test with no look-ahead, plus a live panel on fresh prices.

Why: stay investable, prove the idea on real prices

Module B: Stress test

Event class + impact > 7 triggers factor shocks; loans, bonds, derivatives and equities revalued; P&L reconciles; RAG status.

Why: tie news to balance-sheet loss

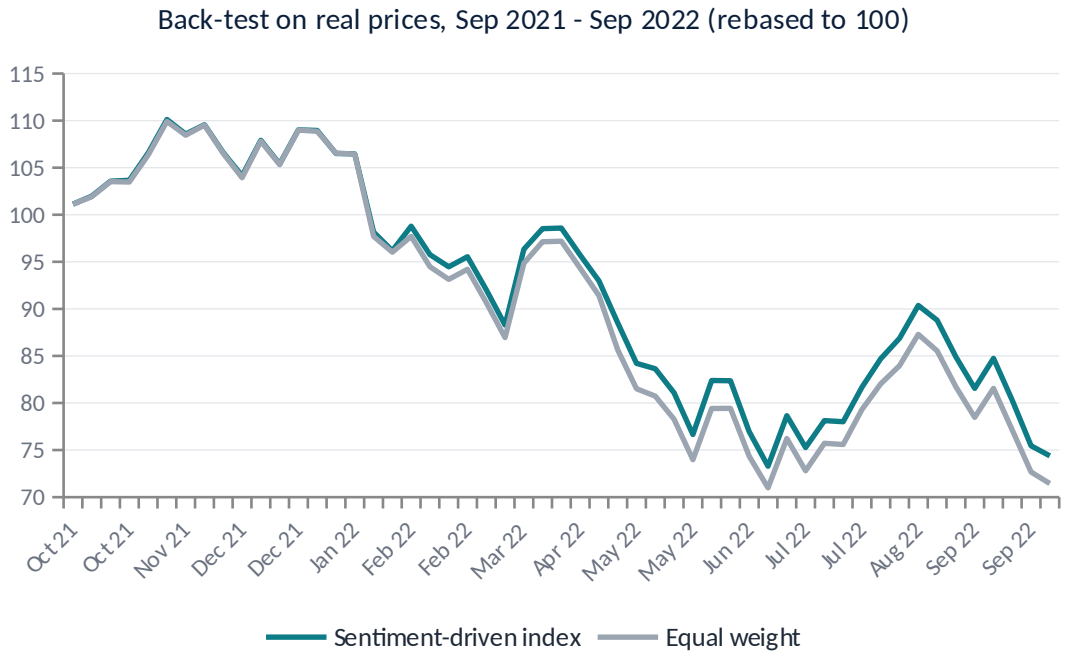
Reliability

Feed fallback chain, retries and 429 handling, batched GDELT queries, de-dup by id and story. 108 offline tests with a mock feed server.

Why: a live demo must not break

Stack: Python, FastAPI + Uvicorn, SQLite, Streamlit + Plotly, pandas, pydantic, yfinance

Key Results



Efficiency vs naive: seconds from headline to weight change, versus manual reading.

-25.62% Sentiment index	**-28.56%** Equal weight
+2.94% Excess return	**0.716** NLP accuracy (baseline 0.594)

Tracking error 2.03% (annualised). The window was a falling market: read as 'lost less', not alpha. Live panel adds forward tracking on fresh prices.

Domain Impact

Index desks

React to headlines in minutes, not hours. Bounds and turnover caps keep automated tilts investable and auditable.

Risk & credit teams

Link a news event directly to loss on the banking book, split by risk factor and asset class, with a RAG status.

The platform

One signal stream, many consumers: new modules plug in by subscribing. Offline by default, LLM optional.

Same signals, consistent and auditable: tactical index view and strategic bank-book view.

Limitations & Next Steps

Limitations

- Tweet sentiment vs next-day return: IC +0.02 (t = 0.7), not statistically significant
- Lexicon NLP is tuned on PhraseBank, so 0.716 accuracy is optimistic
- Impact is a heuristic; stress shocks are judgement, not regulatory scenarios
- First-order valuation: no full repricing, netting or collateral
- Loan book is derived from card data, not real exposures

Next steps

- Fine-tuned FinBERT or LLM classifier for sentiment and event class
- Collect longer live history to test the signal properly
- Regulatory scenarios (e.g. rate, credit and equity shocks)
- Full repricing of derivatives; netting and collateral
- Real-time streaming feed and alerting